

ADTA 5900/5560: Assignments

Important Notes: Being CAREFUL is Being HELPFUL

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1. Overview

To build, train, and test neural networks, the student needs to write Python code in Jupyter Notebook documents. In addition to writing the code, the student also needs to **add extra text** (using Markdown) to make **the code more manageable, legible, understandable, also easier to learn, check, and avoid mistakes**.

Additionally, while **working on the code** of any programming languages, being **careful** is always being **very helpful**!

Any error, no matter how small or trivial it is, may be **very costly** – time and work – **to debug and fix it**.

2. A Costly Trivial Error: An Example

A student of an AI class in a previous semesters wrote the following lines of Python code while working on a homework assignment:

```
[ ] x = tf.placeholder(tf.float32, shape={None, 784})
```

```
[ ] y_true = tf.placeholder(tf.float32, [None, 10])
```

```
[ ] x_image = tf.reshape(x, [-1, 28, 28, 1])
```

3. What happened?

In the above piece of code, he/she made a **trivial mistake** that was **very costly** to him/her – **time and work** – to debug and fix it.

He/she mixed up **{** with **[**.

When he/she ran the code step by step, the flow went through them and all the following steps without any errors until he/she trained the neural network.

It means that the **step** (or the **code**) **where an error was displayed** is **very far** from **where the error really occurred**. It is very hard to detect and debug the error.

4. Recommendation

All students should learn from the above student's mistake and avoid similar mistakes and other mistakes while working on coding in class assignments.